ActInf Livestream #019.0 ~ Deeply Felt Affect: The Emergence of Valence in Deep Active Inference

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hello welcome to the active inference live stream this is active inference live stream number 19.0 on april first two thousand twenty one welcome to the active inference lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at our links here this is recorded in an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome and we'll be following good video etiquette for live streams as outlined in this checklist this is the schedule that we've had so far for 2021 and the upcoming two discussions on april 6th and the 13th will be number 19.1.2 on this paper that we're going to be discussing today so right now in 19.0 the goal is to set the context and give a little background for 19.1 and 19.2 and people who are just learning about the paper which is deeply felt affect the emergence of valence in deep active inference by hesp smith parr allen firsten and ramstead in 2021 and this video is definitely just an introduction to the ideas it's meant to be a two-way on-off ramp from active inference so exposing the active inference community to ideas outside of active inference and people who might be interested in some of the bigger ideas connecting them to the active inference implementations and we have a bunch of sections planned for today in 19 so thanks a lot to stephen and to blue for helping with the slides and for coming on today so maybe we can just um introduce ourselves briefly and then we can go on to the rest of the slides we have prepared so i'm daniel i'm a postdoc in california and i'll pass this stephen hello i'm stephen i'm in toronto i'm doing a practice-based phd around social topographies and community-based development and i'll pass it over to blue hi i'm blue knight i am an independent research consultant based out of new mexico cool well i think this was a fun paper for us to read and we can just go right into what were some of the big points of the paper and the background ideas before we go through the figure and the model which will take up a lot of the time the paper is called deeply felt affect the immersion surveillance and deep active and it was published in the journal neural computation in 2021 and so i'll read the first one and then either of you can give a thought on that in this letter meeting paper we demonstrate that hierarchical deep bayesian networks solved using active inference afford a principled formulation of emotional valence building on both the work mentioned above as well as prior work on other emotional phenomena within the active inference framework yeah so that's just given us that insight into this idea of valence so we'll be speaking about that more but that's uh that's that's an important part of this i'll say the next one our hypothesis is that emotional valence can be formalized as a state of self that is inferred on the basis of fluctuations in the estimated confidence or precision an agent has in her generative model of the world that informs her decisions cool so we're going to be connecting some of the decision-making aspects of agents and inference aspects of agents to emotional balance and they're going to demonstrate it so those are the goals and the claims that they set out in the paper any thoughts blue or do you want to go to big questions and maybe introduce a big question i'll introduce you to the questions all right go for it so the big questions of this paper were how can we make more active inference models or agents that are expressive powerful realistic and interpretable what are the implications of this active formalizes our survival and procreation in terms of a single imperative to minimize the divergence between observed outcomes and phenotypically expected i.e preferred outcomes under a generative model that is fine-tuned over phylogeny and ontogeny thoughts on this well the first one this to me is about just developing the active inference framework to capture more of the patterns that we see in nature or in designed systems so how do we make systems that plan for multiple time steps that's deep active inference how do we make agents that interface with the environment so that's sort of a big move overall and then this is going to be zooming in on the valence and some of those affective aspects that's what on the first point and then the second this is a quote from the paper and it says something that active does it formalizes our survival and procreation not in terms of evolutionary fitness again you could agree or disagree with the authors or think it's consistent with fitness or not but claiming that the imperative is to minimize a specific kind of a divergence and so what are the implications of that if you think okay well if that's the imperative then where's the situation where we've been acting like there's a different imperative and is that consistent or inconsistent with um this and then the last quote and then someone could give a dot is a quote where they say we conclude with a discussion of the implications of this work such as the relationship between implicit metacognition and affect connections to reinforcement learning and future empirical directions so what does active inference or this model say about cognition or metacognition and reinforcement learning yeah if i could add something i think it's it really opens up and a lot of the philosophical questions and starts to give a modeling interpretation so how the body comes in how so the emotion starts to tie in to a way that the body comes in an n-active inference starts to become modelable and it starts to also show how affect is somehow connected um to all cognition and all processes so it's not some weird artifact of shame which might be more of a freudian trap but it's actually an integration of the cybernetic process all the way through in some way very nice it's also captured a little bit on this slide which is recycled from some 2020 vintage active which is asking about how active inference is going to be saying something about the relationship between agents in the world and we're imagining an agent not a one-step tic-tac-toe player but we're thinking about an agent that's imagining through deep time and they have imagination or the capacity to simulate and then they have affect and they can also predict their affect and have precision and so on so here's sort of the drama masks for the positive and negative affect and then there's sort of different world trajectories that are being estimated through time that lead to positive or negative sentiment and that gets around some of the sort of basic considerations like in the paper they say well wouldn't the agent as they meet their demise wouldn't they have precision over their sensory states and the answer is like yes but it's not part of a long-term trajectory that involves desired preferences so by appealing to model nesting some of the sort of single layer not paradoxes but just scenarios that models might fall into like always going with what was known before never going what was known before these can be resolved just with empirical modeling interesting one reason why it matters and we'll come back to this when we quote the authors later is that these kinds of richer models lead more closely to experimentally tractable and unique predictions of active inference because if the only thing the model predicts is the reaction time that might fit the distribution better than some other model now if it can also fit the reaction time and the eeg and also the self-report it's starting to become a very rich multi-modal and multi-measurement way of talking about behavior which is what a good instrument should do so this is helping us come closer through these contact points with other big important areas of research like reinforcement learning and autonomous computer agents and just inference agents of all kinds especially ones with action but then maybe some of these other areas that we'll bring into the discussion because if we can have a scale-free framework that does apply to cognitive agents and then we frame it in terms of the input and output and the actions and the policy the generative model all of these terms maybe some of these other systems like online teams might be able to also be said to have at least computationally affective charge or precision or different things like that so we kind of talked a little bit about that in our 2020 paper for online teams any thoughts on sort of the bigger topics that this paper is up against before we jump into the abstract and the figures and everything if i could just say one one thing i think that's very interesting is that it opens up um the the realities of trying to bring in the mind the body and the environment and there's been this question well how do you do that and why didn't we do that before was partly because we wanted to be able to measure things and lock things down and and the question of how do you bring this whole dynamic in how could you possibly get a handle on all of that um well emotion seems to actually make a lot of sense in terms of or let's say valence should say valence more broadly it makes sense in terms of how it could be calibrating the more conscious levels of that process so i think that has implications for scaling psychology and bringing consciousness back into psychology and helping to really move into a ecological and ecosystem approach to psychology lou any thoughts or do you want to start with the first part of the abstract i'll hold my thoughts i have somebody but i don't want to ruin the surprise of the paper so let's start off on abstract agreed fun times all right um does anybody want to start with the first part of the abstract i'll take it the positive negative access of emotional valence has long been recognized as fundamental to adaptive behavior but its origin and underlying function have largely eluded formal theorizing and computational modeling using deep active inference a hierarchical inference scheme that rests on inverting a model of how sensory data are generated we developed a principal bayesian model of emotional valence this formulation asserts that agents in further valence state based on the expected precision of their action model an internal estimate of overall model fitness which the author's term subjective this index of subjective fitness can be estimated within any environment and exploits the domain general generality of sorry my screen is in the way second order beliefs which are beliefs about beliefs there nice so in this first part of the abstract kind of priming what this whole research agenda is about i'm seeing a few threads one is about fitting good models overall model fitness especially with a subjective twist so that's another kind of a bonus that it's actually not just objective model fitness it's subjective so it's about good fitting models with this recognition that that's like personalized okay so that's cool thread then there's the first words which are about the positive and negative aspect of emotional valence which is despite the richness and the amount of experiences and all the diversity of altered states of consciousness and adjectives there's also kind of this um approach versus recede binary and um this is as they're saying it's fundamental and we see it in living forms like go no go signaling and we also see it in computational agents and so this is putting a lot of different kinds of systems that have very nuanced maybe behavior in their outputs but at a top level they act as if they have a sort of go no-go that in a way helps them be adaptive in their own niche and then the way that they're going to go about that is actually with a second level or second order cybernetics model which is by enabling belief about belief maybe some of these gaps between systems can be bridged so interesting research questions that are not always brought out because the abstract is so reduced and they have to say what they did do but for those who are interested in those questions that's good to know all right i'll read this one and then anyone can give a thought we show how maintaining internal valence representations allow the ensuing effective agent to optimize confidence in active selection action selection preemptively valence representations can in turn be optimized by leveraging the bayes optimal updating term for subjective fitness which we label affective charge or ac ac tracks changes in fitness estimates and lends a sign to otherwise unsigned divergences between predictions and outcomes that's pretty interesting we simulate the resulting affect of inference by subjecting an in-silico in the computer affective agent to a teammate's paradigm requiring context learning followed by context reversal this formulation of affective inference offers a principled account of the link between affect mental action and implicit metacognition it characterizes how a deep biological system can affer its affective state and reduce uncertainty about such inferences through internal action i.e top-down modulation of priors that underwrite confidence so this for me like was a big surprise of the paper like the the valence and um the relationship with valence and subjective fitness so or precision accuracy this was just a really different take like compared to the last paper that we did just like on valence was very much like positive or negative in terms of like feeling or emotion where this is like really positive or negative just based on like surprise essentially like what if my model doesn't match my expectations like or or the outcomes if my expectations don't match my outcomes that is negative valence which that was a kind of surprising take on this for me compared to the last valence paper that we did and the other thing is really this the way that they've modeled this is totally like a perfect example as they talk about in the discussion of the bayesian brain and why bayesian statistics and updating um like especially when they talk well i'll hold i'll hold my thought for them for that time but but really this was just an exemplary paper for that for me nice yeah one one thing that came up for me there is it talks about you know representations again and those so this idea that internal valence representation so suddenly we've got a way to bring representations back into um what had become quite a non-representational modeling approach and that's really interesting because it's like normally you think of representations previously like the most concrete is the thought then there's the body and there was this fuzzy thing called emotions which somehow was there and axiology and all this stuff but now it's like well wait you've got an internal valence representation which could inform the body and the cybernetic thinking metacognition process so it it gives a way in for representations but it's not representations as we normally maybe approach that term so that's quite cool all right third abstract who would like to read it okay i'll read this one thus we demonstrate the potential of active inference to provide a formal and computationally tractable account of affect our demonstration of the face validity and potential utility of this formulation represents the first step within a larger research program next this model can be leveraged to test the hypothesized role of valence by fitting the model to behavioral and neuronal responses nice so i set up the problem by being about using higher order cognition this second level modeling to approach some of the interesting dynamics of agents like their apparent affective charge influencing their behavioral decisions in the present moment based upon inference about the future and they're going to demonstrate a simulation there's going to be kind of a model layer and then there's going to be a simulation that goes down into the t maze and a mouse and so there's sort of two levels of generality that we're going to go through here they start the paper with the qualitative but still rigorous level of generality which is just talking about action in different systems and multi-scale biology and the need to integrate bayesian approaches into how we think about evolution and development ecology and so on and then they turn directly to the methods and perform four-step walkthrough that will be somewhat familiar to those who have seen model stream one with ryan smith and christopher white where they also step through the construction of this model but it's something that as we'll probably see soon it's it's just always worth it to kind of build up this model incrementally then they with this implicit metacognitive model at step four of their models elaboration they discuss a few kinds of important key terms that are like bridges between previous computational psychological models and their active inference model and then they implement a simulation with a rat in a team ace which we'll discuss as well and then they bring up all these very interesting questions towards the discussion section of the paper some of which we've exerted and conclude with a discussion on the future empirical directions so before we go into the affect and model simulation implications any thoughts on the roadmap all right let's go to affect and valence this is quoting from their paper so definitely read it to see everything that they say but they refer to classical constructs of valence so maybe here just representing the classical valence construct on a kind of clock chart what would either of you think or say about valence or what motivates the study of valence this way or overall so i just had a comment and it's based on something that uh the authors had said about surveillance so here uh we've got the arousal and like high and low arousal on one axis and positive and negative on the other axis but the authors had mentioned that uh some people see valence as like a an array of positive like it's a positive and negative or two separate axes of valence and i thought that that was very interesting whereas like like it's a it's not a um you know a binary choice positive and negative but there's a range of positive and there's a range of negative and ambivalence like contains some degree of both i thought that that was an interesting uh take on valence i've thought about before cool and they bring up good scholarship with bringing up a uni-dimensional view so that would be like yes there's like a thermometer and it's kind of just like it goes up or down it's a number that has to be strictly ordered and less than greater than maybe even linearly correlated with a hormone or a chemical or a neural firing in some region or a pattern of activity then maybe there's multiple dimensions maybe there's um like as here an arousal and a high to low arousal and high to low valence positive negative um maybe it's multi-dimensional so what's cool about the active inference framework is we could test the computational roles of these kinds of variables and ask okay this is fitting the data better but is it um more parameters is this worth the trade-off that we're getting here so we're kind of taking this single dimensional model and thinking about it with a family of other models where we could test the influence of other factors and see if such patterns are detectable in the data or ask what kind of behavioral settings or groups or questionnaires or whatever it might be would elicit um a differentiation on another hypothesized axis yeah and just one thing that goes with that is it sort of shows you the current ways to get at this is kind of through self-reported emotions and using this kind of graph and actually use this a tool with our community with disabilities for um reporting back on their experiences but you you're essentially still getting this emotion um as as a self-report and trying to track back um using sort of assumptions about how that plots with other self-reported emotions and um whereas the the and this is also used a lot in uh market research for marketing and advertising this tool so but actually in this approach here they've got a way to access this without just getting self-reported words um so this is this this i think shows how it has you know this this is an area that's got a lot of application potentially cool here is a slide just upon uh differentiation in english between affect and effect so not to split hairs or to introduce words that they didn't use in the paper like effect but what would be the point on this stephen well you've you've got just to give the sense that it is often confusing those two terms so if we have the person these two people on the left so you've got um the person x has experiences and effect some sort of you know relationship to person wine it's it's created some affect now that could be anger if it was reported later and they said what was their emotional state and this affect contributes to a policy selection where person x interacts with person y and pushes them into the pool so you can see that okay so that that um the effect of this is that that person y ends up in the pool so there's there's a there's a there's an act and there's a cause effect type of um linear type of mechanical action that happens um and of course then there can be additional affects on both the parties and effect resulting from this particularly if the person was to uh you know drown or something depending on the scenario so i think it shows that there's a kind of one's more dynamical and relational and one is more objective i suppose you could say and cause and effect related sounds good and here is from a previous um discussion i think number active 11 where we talked also about and about trajectories through affect space so whether or not it's like how things quote really are it might be a behavioral index or summary statistics of of someone's online activity even if you're not directly measuring them or getting their self-report and there can be trajectories through this space now one topic that um a sort of detail that's a little bit special for active inference that doesn't get emphasized a ton is that basically the certainty in the model is one and the same thing as having better better affective charge increasing certainty and high certainty in the are taking the place in a reward centric of highest expected for example value and it's one major difference between um an active inference approach to thinking about economics or decision making versus a sort of just like classical um maximum likelihood reward maximization so in these active inference frameworks they're not modeling a happy to sad variable and calling that affective charge they're actually modeling a precision variable and then saying that when things are more precise and are getting more precise there is low negative affect so it's kind of like positive and then anxiety like negative arises as a function of uncertainty in the model and um here's just some words and some thoughts that we had put on anxiety from a previous one but any thoughts on this about the sort of integration of valence and certainty rather than valence and expected average what that means or how that's different than other models that others have seen yeah well i think that does have quite important implications because it's now prospective it's kind of what was the contextual expectation on where things would go um as opposed to just it being this kind of static expectation on the moment itself and it's all collapsed down into sort of some sort of reductive there's the idea that there's the person's valence is dynamic in relation to um you know their expectation but then there's also the precision at which that expectation then resolved or didn't resolve you know so you may expect things to go badly and then things went worse than you expected or it went better than you expected but it still might have gone badly but it didn't go as bad so it gives you much more um well i don't know i wouldn't use the word realistic i don't know what term you'd use for that no yeah that question cool blue so i can't remember which paper it was exactly but it was talking about um anxiety it was one that we did on the live stream talking about anxiety as like a fixation on the future right and i talked about depression as a fixation on the past was it the big five i feel like it was the big five paper um but but that is just an interesting um and kind of different perspective on anxiety as as relative to the valence that's presented here cool and uh the question of how to map model parameters to experiences that we have which are encultured and which are relating to our individual kind of qualia that's a big jump too so we shouldn't just instantly also paper over the computational psychiatric uh gap but that's what we're going to unpack is what does the model really say and then what is that going to help us do or rethink one note is that they wrote that they are referring to their model distincting it from just sort of active inference more broadly as affective inference because it sort of has that the active in there but it also has affect in there and they're saying there's what makes it different than just general active inference and uh one of the one attribute that they apply here is deep so that's meaning deep through time or potentially also using deep uh learning like deep neural network approaches we had i think christopher white or someone it all blurs together but somebody gave a few senses of deep active inference and then they're also modeling affect and so that's why they call it affective inference so it's deep affective so if you know what active inference kind of is about they're taking that skeletal framework and extending it through time and adding an affective parameter or um a set of interacting variables way to think about and model the kinds of things that people study affect in that's what they want to study with this new development so that's kind of what i think they're doing in this paper and um it's also a little more general than just affect because as they say it can generalize to potentially other higher level explanations for conditional dependencies between higher level parameters and how they reflect behavioral outcomes so that's a pretty general statement that's very important though it's saying that it could be possible to have other higher hidden states that cognitive ideas let's say and those don't have to be just the positive or negative approach or recede that they model here it could be something that's a lot more subtle or symbolic or categorical rather than this is more of like foraging go no go so this is a initial foray into this kind of deep and hyper parameter reaching back down affective in that sense and it's tied to the affect rather than to the for example cognitive work but it's as they say implicitly metacognitive and stephen as you said it's not that affect is late in late late and then like strikes into the scene it's actually that there's this sort of fundamental cybernetic role for affect regulation so it's not a red flag mode that is explosive or something it's something that's intrinsic like metabolism okay they uh bring up a lot of previous work and some limitations of the model so check out the paper more if you want to learn about for example if you think well that's a very narrow uh model definition of valence they might bring it up in their paper so check that out but let's just get right into what the model is and there's going to be sort of two acts coming up here there's going to be the model which is just the mathematics of what they lay out and the formulisms they lay out and then we're going to go into the simulation with the mouse or the rat or the ant or whatever in the t mace so the first is going to be general and the second is going to go into a simulation that they're going to get results from as well so we're going to walk through kind of four clicks in the elaboration incrementally as they write on an active inference model okay so fun stuff thanks everyone also joining live so here we are in figure one who would like to go first maybe blue you annotated this really well so what did you see in figure one with model one and model two sure so model one is um the generative model of perception which we've seen before where s is hidden state o stands for observations d is prior belief uh a is this likelihood mapping or like a probability distribution and then tau is time which we've seen before so in m1 there's no time this is the first step of a generative model and it's just perception so it's it's like sensory input and the posterior oven over hidden states is all that's really taken into the model of perception and then the m2 so as the models as we go through these m1 m2 m3 and m4 in the next few slides there's an increasing layer of complexity as we transition from m1 to m2 to m3 and then m4 with m4 being the most complex and so in m2 which is the generative model of anticipation the idea of perception is then extended forward and backward into the past if you want to go there using a transition matrix which is this b sub t for to represent the hidden state so it's just an extension of of perception essentially and it includes the anticipation what you expect to happen as opposed to the just the sensory input which is just what's present in m1 so it's anticipating hidden states over time m2 nice awesome good explanation and we can look at some of their or anything else to add actually or ask here stephen on figure one was really helpful i mean the only thing i might add is that as you were mentioning that is it's almost like the the predicted input in that funny sort of way it's like so we've got what was what's the predictions of what the sensory input is if i could sort of somehow infer that and that that's that dynamical so i i think this is really helpful though to go through it this way yep and to give a note on that a lot of times when people think statistics they think descriptive like we're going to take the big data set the big you know image data set and we're going to describe it with a summary statistic and that does map onto this topology like observations get mapped through likelihood onto the most optimal state of the world but where generative bayesian modeling comes into play is that that model that's inferred that hidden state what is the real temperature or what is the real average height of the classroom it can also be used to generate and to combine with other models to generate testable distinctions you can generate data and then this duality where you can like take a little bit or a ton of data and then update your prior already using the bayesian terminology and then use those prior estimates to generate more that is like expectation maximization or em optimization or the parametric empirical bayes approach so it's not just taking big data and crunching it into a regression coordinate and saying all right we found the most likely one it's done this is actually a framework that allows us to bounce back and forth between summaries of big data and the actual large observations themselves and then here you can think of the prediction is on a single observation with a single hidden state and here it's like getting a vector through time it's then you can imagine that it would be more dimensions but this is just the skeleton so let's keep on walking through yep let me interject here really quick sorry so can you just go back there we go i love this what the authors how they describe perception i thought it was very clear they said perception corresponds to a process of inferring which hidden states or posterior expectations provide the best explanation for the observed outcomes i thought that that was very clear cool and the soft max is the sigma and that's just saying it's like a decision-making criterion uh function but we can also ask the authors more because that's sort of the machine learning uh term yeah so just one just one very quick thing on that as well i like what what blues just said there as well is by bringing it on the outcomes as opposed to that that outcomes brings in action so you you you because the out it's it's like the perception it's not what the data is telling me it's also the outcomes of what happened in the world with the body and that that is really quite i don't know if that's been so easy to bring in before um in the language so i thought that's quite interesting yep and we're going to bring policy in in a second but for now you're right we have the difference between the outcomes and the hidden states is it raining or not that's the hidden state and then is my shirt wet or not that's the you know sensory state but we have why is measured wet right is my shirt wet yeah exactly but we haven't gotten to this policy level which is should i go outside so table one is just describing in a little bit more detail the p and the q distributions that they use um in their figure one for this like basic level and that uh comports their math with just the broader modeling bayesian statistics letters used although the a b c d are sort of specific for active so let's go from uh m2 model and bring in policy so this is m3 model this is the next click that actually gets us to the first thing that looks more like a real active inference model beyond just a bayesian time series approach so maybe blue go for it again with what you would say for m3 so for m3 this is the third step and this is the generative model of action so here it's inferring in phenotype congruent policies from all of the terms that we had talked about in m1 m2 and then this brings in policies which that's pi policies and then e sub pi is the baseline prior over the policies and g sub pi is expected free energy c is phenotype congruent preferences and then f sub pi is the variational free energy conditioned on a policy so the policies here already here and we just have these uh m2 is the what's perceived over hidden states uh based on perception m1 so this is kind of just adding that that ex extra layer including policy into the into the action model yep so it says here posterior policy beliefs are informed by the fit between anticipated and preferred outcomes while at the same time minimizing their ambiguity so to minimize that so nice thanks for this a connection between variables in this kind of a graph represents them having like a statistical dependency so we can look at pi which is policy and then ask okay so we had m2 which was this time series that we that looked kind of like a i don't know like a rake or something or a horse and then paul policy pi gets linked in not directly with the thumb on the scale influencing the hidden state but actually influencing b sub t which is the transition matrix through time of that hidden state so one thing because there's no pi onto the hidden state you can't have policy at a snapshot like timeless policy all policy is intrinsically related to the movement of because it's hitting the b matrix not because it's not plugging into a hidden state directly so you're always talking about a temporal model just how deep are you talking and then it also conditions the estimate of how the hidden state evolves that's what that vertical line like given you know a given b so this is we're going to do the time series how how much um bitcoin will i have under the policy of not acquiring any zero easy estimate easy heuristic so then those kinds of conditionalized estimates of different kinds of things can help agents act in the world so that's why policy is interesting to think about and so instead of thinking about how to optimize a massively complicated scheme for maximizing utility with an unknown variability this is allowing the agent to model how one is embedded in hidden states estimate or changing through time so that's the pi part and then also this is kind of explore exploit so minimizing the ambiguity is one imperative but also you want it to be fitting the whole time like i want my shoe to fit but i also want to figure out where it doesn't fit but i need to be walking the whole time and so by keeping a bunch of these features linked in the model the dials at least exist to modulate in really creative ways like e is this baseline prior and so you could modulate it so that the is relatively unperturbed by sensory or you could make it very sensitive to specific kinds of sensory outcomes so this is again like a modeling skeleton that potentially could be used to evaluate how different sorts of behavioral phenomena like approach and recede or something like that or freeze whatever it happens to be might be linked by the organism seeking precision on an estimate of how things are going to change in the niche given its policy so of course there's a lot to say because that's kind of an interesting area but that's just the third level so we're integrating policy decision into how states are changing through time and then policy decision is the interaction of several variables the baseline prior like the affordances the habit as some have said it here and then c the preferences phenotype congruent is like if if i'm a shoe size 10 i want a shoe size 10. then if you have a phenotypic preference that's different than what your actual niche or needs are then it will be ill-fit so you won't be fit for your so then the niche will be filled by those that are fit to it and then f of pi is similar like it's um calculating variational free energy which we can definitely ask the authors more about and see how that is related to calculation on a specific policy you know yeah okay any thoughts either of you well just one thing i think is quite helpful as well um is they they call that perceptual evidence uh the the variational free energy conditioned on a policy so it's like um and then that gives you the sort of perceptual evidence and then the actual action model which is kind of like the free energy and that helps sort of um show how free energy is coming in at this kind of larger time scale um and i suppose one thing just to mention to the viewers there is a whole load of nested free energy processes going into all of this process but at this scale that we're talking about it's so that we can understand in terms of policies that we're trying to understand sounds good and so yeah here you know um anyone who has small children knows that their feet grow very fast right so when we're talking about the action model expected free energy it's it's the expected free energy because i don't like the actual free energy can't really be measured in in this situation so we're dealing with expectation so there's this phenotypic risk component on the left side of the equation like how much money am i going to spend these shoes and minus the the perceptual ambiguity like i think uh my kid is a size 13 but they might be a one or two how expensive are these shoes do i want to buy shoes that fit them right now or do i want to buy shoes that they can wear two pairs of socks with and wear for a little while so you're always weighing out this this risk and the ambiguity that that occurs okay in that specific example the specific policies are like okay i could ever measure the feet every day and that's my sensory outcome the hidden state is the true size of the foot and then every day my policy could be if they're in a new shoe size we change that day another policy with a different trade-off structure different measurement requirements might be every two shoe sizes another one might be when the perceived ill fitness is too high the squeaky wheel gets the grease then we have an action-oriented response policy and i'm going to be thinking about all in this model and about the certainty on it so nice yeah i mean it also relates to that when my daughter has a big toe starts to get sore that's my perceptual evidence it's like okay it's like that shoe is rubbing a bit now so yeah at some point i've got to make a choice so that's a good that's a good example actually it's an error signal that it's ill fit into the policy and that's a niche modification with a shoe um but it's a similar idea so table three kind of goes through m3 in a little bit more detail and that brings us to figure three which is the fourth click of the active inference uh kind of chain here and it adds just two more letters it's mostly the same as figure two like the c g and the e which connect at pi that was my mistake in the 18 2 and i connected the e to the um g instead of the e to the pi um they're connected at the pi as shown here um the c g and e are now in m3 with a blue box and then m4 one more dependency is being clicked in here which is gamma which is being clicked to g so gamma is now an input to like a bigger g function so the each the g is is a function um which you know you could input just a number you could input a few depending on how you write it so this is a slightly more expressive g function because it includes this variable and gamma and b are kind of like one over each other the reason why it's kind of like when people take logs or do one one over or negative this it's just sometimes you want to get the most of something but it's easier to frame it as getting the least of something or vice versa um but what this gamma represents is the expected precision because um this is kind of certainty so this one is uncertainty so what would either you say about this final click to m4 blue or steve so this is so this is so fun so the gamma is like how much did that shoe not fit and the beta is like how fast are their feet growing right so i think that that's the i mean i guess beta doesn't take into consideration time but it was um it was described as a rate um it was described as a rate so maybe it does include time i'm not sure i don't want to go too too far down the rabbit hole with the shoes but it's almost like there's more precise shoes like you'd be more precise with the steel toe and then you could have the more stretchy no no no no one take this too seriously it's about what it actually does say which is basically the um precision which is the expected precision is in the gamma so that the uncertainty can be yeah in the b and this figure three the reason why they even went to this level of incrementing it out first off is to show us level by level how we're building on okay it's about a time series that's being predicted through time of sensory that are like generated from an underlying time series of hidden states vis-a-vis a likelihood mapping and a transition matrix and that's getting played into by priors at that s1 level at the first state level and then there's policy that's the whole cybernetics control theory and then this is the machinery of the decision making and now we're just adding one more level which is the implicit so now we're adding this implicit kind of always on always being integrated into the model again not a disaster mode recovery mode for the otherwise non-affective agent but it's an affective agent whose implicit metacognition its kind of key regulator here is related to its certitude not the thermometer's tracking the average of um it's its returns on investment or its average estimate but its uncertainty in its generative so that's what we get to with their figure three with a fourth click so wait wait i think we need to talk about here uh because it's the first time that it comes up is there um affective charge here oh yes so in this in this model and which also m4 really is the top level illustrating like the hierarchical um hierarchical nature or the nested nature of how this model is is normally and then in the effective charge it's it's described here as phenotypic phenotypic progress which is the policy in the prior minus the policy in the posterior and then the free the free energy of the prior posterior minus the prior wait no i said that wrong prior minus the posterior there it is yep here here it is in table four so let's look at what they say so expected precision is gamma okay looks like a little dart gamma you want more precision in an optimistic world model and policy pi okay and it turns out that gamma the expected precision is one over a gamma distribution that's gonna be the beta and that's just gonna be a statistical distribution that plays nicely with statistics just like you see like natural log like the gamma the normal these are tools that statisticians use and so they kind of work here but we can ask the authors to what extent does it have to be this way but anyways gamma is the expected precision so gamma and beta we can kind of think about them as like linked because again they're one over each other so wherever you see a gamma or beta it could just be replaced with the other and affective charge is being um defined as yeah the the uh change in the multiplied by the free energy of that so not just if if it's good but unchanging then this is gonna cancel out so the phenotypic progress is low if you expect you know the 10 bitcoin and you see them and you're not changing your policy in formulation that's why we don't need to tie it too closely to a specific emotional experiences because those are going to always be super subjective and require a ton more specification but we're just kind of broadly calling this ac variable something that's related to both the way that the phenotype is changing the policy is changing and the way that the free energy estimating of that policy is changing but there's something important to note in that equation is if the policy doesn't change then the affect of charge is zero yes and let's again let's have the authors give us the authoritative um uh information on how to read some of these terms but here is what they said about that step four the fourth level which is again this gamma precision is required to enable the asian to estimate its own success which could be thought of a minimal form of implicit non-reportable metacognition so even to be able to semantically express metacognition like i am unsure that sentence is many levels or another level away from just this precision variable which is like an in it doesn't have a self report circuit this is just like a model that has an error term that it's updating at a slightly higher level than just the likelihood mapping and the expected precision term that's gamma reflects prior confidence in the expected free energy over policies the expected precision term modulates the influence of expected free energy and policy selection relative to the fixed form policy prior e sub pi which is the habits or that the so it's kind of like should i fall back to priors or should i go with what the g function lean into what my g is telling me then um yeah formulated this way we can think of of gamma the precision as an internal estimate of model fitness subjective so higher precision my internal model is doing better lower precision my internal model is doing relatively worse than it could be it represents an estimate of confidence in m4 that fourth click in the phenotype congruent model of actions given the inferred hidden states of the environment so they kind of tessellate it back out there they're saying it's about those outcomes and about what generates them and how those change and how our policies play into it and it's about how certain we are in that generative model and there's a lot of ways the environment can change so if we're getting good data from our observation then we should be reasonably confident with staying with what works to some extent but if that should change there should be another mode that allows us to rebalance preferences affordances um and integrate that in a different way in a different allocation to our policies and then to emphasize its relation to valence in our formulation remember valence is like feeling good or bad positive negative going forward we refer to gamma updates using this term affective charge so here's ac affective charges change in beta aka change and gamma in a just another way of saying it it's change in and then you can imagine if if the precision goes up it's going to be good precision going down bad and flip it for the one over or however it ends up being written but that's what it's getting at when the precision is going up as new data are coming in like they're you know i'm thinking of a slot machine and then it's like oh you won something and now the second digit is going to spin so there's both an increase in charge and then that like kind of restabilizes and now it's about whether that changes again or whether the second digit is better and so that is how affective charge is being tracked through time with these gamma updates okay any thoughts or continue 32 well i think well i think the one thing that sort of i find helpful here in this here is it's it's the general overarching active um parameter of uncertainty is like it's it's they're using the term confidence and precision more at these higher levels because it's like a meta intern is still at some level uh an interpretation of uncertainty but because it's on policies in that confidence and precision are at play because you're interpreting your ability to act not just the raw sensory sort of mellow so to speak right there's a big difference between like visual uncertainty it's just two shades of grey and they're so similar you just don't know what you don't even see a difference versus two strategies two chess moves that are very different and you're just not sure which one to do that's a very different situation for because the way that your eye scans on the page is kind of reversible you get you're getting to retract your your choices but with strategy it might be very irreversible very non-linear um okay here's another example they give on 32. so this carrying on right below equation 3.1 which was screenshot there so um ac can only be non-zero when the inferred policies are different from the expected policies so blue that's what you had said so if it's sort of at the bottom of the bowl from a policy perspective thermodynamically it's not moving the delta is zero it's positive when perceptual agents when it is positive when perceptual evidence favors an agent's action model and negative otherwise so it's like are the data coming in locally good for my model or locally not congruent with my model in other words positive and negative ac corresponds respectively to increase and decrease confidence in one's action model so as the data are increasingly good you actually like lean into your model and prioritize that prediction and then you kind of start maybe overfitting and then maybe the model has to pull back and rely a little bit on something to to re-equilibrate that to stay on this multi-dimensional front with model fit and optimal learning so that's one of the ways that active inference is rethinking the explore exploit is by relying on the generative model and then the trade-off with different ways the generative model can get combined with priors you can get explore and exploit behavior as well as a lot of other modes and a rapid switching that's informed by the instantaneous performance of a um then they give some examples of different a prey and a predator animal okay and then here's three any yeah any thoughts on that before five three okay five three affective charge lies in the mind of the beholder so that's cool because it shows that the authors in this research line have taken up the challenge of ecological psychology or of action-oriented philosophy or whatever it happens to be it's like this relational insight that these variables have to be about the beholder and their relationship to a situation partially for computational tractability and partially just because it's kind of how these systems seem to be we are modeling things that are in the mind of the beholder so it's always going to be with the situation and the context and we're always going to have to go in to the specifics um but we're also going to find the patterns that are similar for example we all as they say tend to experience a rush of satisfaction not five hours of satisfaction unless it was a long jigsaw when we solve a puzzle or understand the punchline of a joke or act of inference an aha moment our explanation is straightforward in active inference biologically plausible forms of cognition inevitably involve policy selection policy as fundamental so policy is not the disaster mode this is as fundamental as affect whether those policy internal selections the selections are internal citation citation citation or external period therefore ac is also elicited by mental action typically in form of top down modulation of lower level priors i'm getting the lower level pain signal i shan't not think of it and then now i'm feeling better as that is top down prior is influencing the bottom-up signals just to give one so this is like bringing a lot of pretty interesting areas together especially because again it's not just about humans as cognitive agents this is a scale-free formulation for how affective charge works in active but we can look at the specific example that they do provide with the simulation so blue any thoughts here or on figure five as far as what they do in the simulation um maybe should we go through the simulation first do you have that figure up or no yes i think it's is it this one yeah figure four okay sorry yeah yeah i think that yeah that's probably better um so in this simulation the way that they set up the teammates is um it's a rat or it could be i mean it's a simulated agent right so it could be a router it could be an ant or it could be anything in a maze where on one side of the maze is food or so the mouse is placed in the middle of the maze and on the north end say you have on the left side food and on the right side a shock and on the south end there is an informational cue about whether the or about which side the food is on as opposed to the shock and so the food location changes over the trials uh and then the um the subject is allowed to move themselves and so typically the subject checks the uh queue and then gets the reward so there's like some there's a uh and a reward for like you don't get shocked accidentally if you first go see which side that the food is on so how the cue is associated how the cue is associated with decision making well if if they know to epistemically forage like that that's more valuable than you know accidentally getting shocked then they go to do that but i think in the trials they changed they kept the location of food consistent for many trials and then they changed it for for many trials and so that the mouse had to relearn what to do yep so let's just imagine the food was always on the left side never changed well as soon as you went to the left side 10 times you'd be like i don't even need that q anymore it's kind of useless info gain it's going to be not good time spent so i should just go for the policy of going right for the food now if it was switching every time it would make sense every time if the shock was painful to get that cue every time so the mouse is able to in one way make a binary decision which arm to follow for the food but also there's this higher order decision about getting information and actually in the active infra infrarents the ant paper that i did with some of these um co-authors we also used a tea maze but there was no metacognition there was just the stigmargy of the colony so the teammates is really a generalizable framework and once you have the code to make the teammates you can just make basically anything it's just like any other shape maze is just another matrix so this is again it's like an initial foray and the places where people see that it's lacking like that's where we want to build in the gaps let's get to a little more formal view of how you make that active infrarat that is doing the act of inference on this foraging decision so this is a there's a bunch of sections to this figure and blue would you say it's better yeah i think five is better before seven right yeah okay so there is um a generative model uh first off i mean there's a lot of entry points people should check out the model and just read it themselves because it will take there's a lot of ways to enter into it but one thing to notice is this c the preference over outcomes if it's not if it's food it has a plus four points shock is negative six these are just numbers you can make it scaled differently and then if it's not a shock or food it's zero and then also it doesn't have a preference for going left or right so these you can already see are some of like the little atoms that are going to get fed into a bigger so this bigger model is going to be able to take kind of these variables like how much is the relative weighting of the food versus the um lightning bolt or is there an implicit preference for one side versus another and then here the prior d the priors that they're 50 50 that it's on either side but again that's actually different than a preference for left or right branch is different than a belief that it's likely to be on the left or right branch so we can separate out those two variables i wish it were this temperature that's my preference but i'm very accurate that it's not that and that's why i'm sad so then you can talk about those kinds of combinations of different parts of the model and uh b so that's d and c b is the state transitions and so these it's kind of like a second level matrix and this is kind of it's it's like a cube if you think about it's like a four by four by four because there's four positions it can be in the waiting position that's this front then it can go to left to the right or to the bottom and then there are state transitions that are defined like if you go from the center to the to the left branch that's going to be a defined move in a matrix that you can make then a is the mapping between where you're at and whether you get the cue left or right so that's the info that you'd get on the bottom the south branch and then the reward or the punishment and that's the likelihood mapping whether you go to the left or the right regardless of whether you got the q or not and then e is that baseline prior over policies which um is just it's even here they're all 2.3 we could ask what is the effect of different values it might make you know you could work that through in your head a little bit but it could be different but here all of the expressivity is in the other parts like the preference for the um reward over the shock so one doesn't need to specify like yes and i'd rather you know take a left turn at this exit while i'm going there but that could be specified because again it's just sort of a um skeletal model of this scenario any thoughts on this before we go to seven nice but isn't it it's cool model and it's nice how even with only four locations there's so much richness in the behavior so shows you don't need the little scurrying animal to simulate the spatial dynamics of the maze because then you're going to get trapped up oh is it a narrow chamber or a broad one but this helps you look more semantically perhaps at the way that the agents interact with their environment okay so now on seven um blue do you want to say the two halves of seven yeah so this uh part of seven this is the generative model that they've created for aspect of inference and the next slide is going to talk about all of the different uh matrices that go into producing this model so that's why i don't know maybe it's this one maybe this is better first go forward um so the uh here the the different matrices that go into the the larger model in figure seven include the prior expectations d2 for any initial states so that's this uh top left and then the second one which is down here on the bottom is the affect of prior which is the likelihood matrix reflecting some degree of uncertainty in the predictions which was also shown in figure 5 multiplied by that beta and sets the expectation on expected sets the expected precision so which varies between negative point five which would be a negative um a valence and then uh two which is positive valence uh and then this contextual prior is this matrix aac a to the c for the likelihood mapping from the context states to the lower level uh reflecting the fact that the agent is always certain which context they observed after the trial is over so they either got the food or the shock and they know that right and then this last one is the state transitions b b2 b to the two um and this is state transitions at the second level reflecting a couple of assumptions that both the affective and contextual straight states vary but have some stability across trials 0.2 to 0.3 probability of change and that the agent has a positivity bias in the sense that they are more likely to switch from a negative to a positive state than vice versa thank you blue so what's happening in seven is we're adding um another level so here we see that beta and gamma being connected to the whole cge policy and then that rake formation and now we're going to tie in this implicit metacognitive layer with another hidden state so that introduces this parentheses notation where if there's a parenthesis it means which level of the nested markov decision process it so that the s that we knew and loved that was giving rise to the sensory outcomes is now s one and that's the first level of the model the s that is given rise to the precision is s2 the hidden state so that is now what is being called the implicit metacognitive component which is the second level of the model s upper score two which has its own prior d upper two just like this has a d upper it's actually not shown here but it is there um there are uh the same markov decision process structure is being recapitulated but there's a different interpretation because the hidden states at one level are now the observations linked through an a at the next level so here's the observations like what square am i on the sensory outcome am i and then a hidden state what kind of square am i on and then that's getting mapped to am i happy or what that's the whole question of this paper is what does this actual second state yep stephen oh yep now yeah i'm gonna i'm gonna have to head out sorry thank you peace thanks a lot yeah thank you bye so then you get all these kind of interesting uh different kinds of matrices at the second level like given that i went left and the food was left then will i find out about that and so you need a few matrices to kind of close the doors and make the model run but that's a little bit on the implementation details and we'll probably learn from the authors and in 10 we kind of just have a summary this is i think a great figure because it shows how multiple time scales of nested processes are linked from the inheritance of prior phenotypes like how big is that foot and also development learning like how fast does it change get linked in through multiple levels that are getting nested closer and closer to action and perception in the loop and then rippling back out so pretty nice overarching figure any thoughts on this or continue on blue this just really like like like you said it's a nice figure a good summary and just reflects like the deep temporal like nature of this model that they've built next let's uh go through the results of the simulation and then just deal with a few little further implications and then that will be a nice 19. so in figure 6 they simulate out that scenario that we just talked through a few times with the maze for 32 trials with food located on the left side so this was just testing that it will converge to expecting the left side this was kind of like a little sanity check for their model but also it shows that we're doing this in the in the broader spm toolkit and so that means that a lot of these things like the um ability to visualize and easily run these models and check that they're complete and debug them and run them at scale a lot of these things that tie up a lot of innovating innovative modeling efforts a lot of those are simplified somewhat and again we'll ask the authors about this so they simulated in figure 6 just to show that they could do different policies from the exploratory policy in the first 12 and then switching over to just the go left policy okay figure eight anything to say blue no okay yep so figure eight they're kind of just giving a few trajectories of what can happen so it's it's characterizing the model in a little bit of a playful way but i think it's fair and as we'll see it's it's at least computationally warranted so here we have initially anxious the state is negative for the first eight time steps whereas getting the q and the left side and then it develops confidence and then the green line here in eight is where the um the situation changes so then there's a few time steps where it still hasn't caught on before it gets shocked and then it gets anxious again and it gets renewed confidence so this is capturing a few of the just like with the ants how we showed how it could switch arms here they're showing not just that it that that it can switch arms from q and left left and stuck then the shock unsticks it it's like whoa i was very sure that going left was the right decision i got shocked now i need to go take the q and not just one time i'm going to get the cue multiple times okay i think that i have renewed confidence in this direction so it's really nice model behavior yeah what do you think blue so they did actually there's one thing and so it's not showing here maybe it's gonna show in figure nine but maybe they um maybe they i don't know if they showed it in the figure or maybe we'll just let the authors talk about it yeah so about the model that lacked the the last level the metacognition about like in one of the models they took that that highest level of integration away um to see if the mouse would similarly adapt so here they changed the position of the food right in this in the top right figure so they changed the position of food it was always on the left always on the left always on the left the mouse adapted it didn't even go look for the cue and then in the the model without that metacognition the model thought the metacognition didn't adapt to going to look for the queue like they just like stayed stuck in the middle like not sure do i i don't know if i should go up or go down i thought that that was really like funny so so this is what happened with the metacognition initially anxious developed confidence anxious again and then renewed confidence but in the model that left uh that integration that metacognition it didn't it just like stayed stuck in the middle i thought that that was a very like cool anomalous effect and like a really neat control like trial for for this um test nice and it actually relates again to this idea that affect or anxiety is in the loop and it's something that's a continuous variable and in uh you know healthy settings we could say maybe anxiety is actually warranted here there was a shock in a strategy that was pre there's nothing wrong with a strategy that works and there's nothing wrong with switching strategies when it doesn't work but there is some variable whether we associate it with suffering again that's a second question but there is some feeling or some computational need for the system to actually pull back and re-weight how it uses its prior evidence versus how it deals with incoming evidence so i think this is a really nice example of just the categorical the discrete kinds of behavior that are very different from low precision and negative valence and then a kind of phase transition rapidly into a stable phase into negative and back again and integrating it with decision making with as they say early on the imperative is that reduction of uncertainty with the not maximize reward not minimize suffering either but actually to optimize its precision aka reduce its uncertainty about a generative model of the states of the world including its own action so it's just really nice work some of the reasons why active inference is so cool actually and then figure 9 they simulate 64 trials oh i think this is what you were speaking to an affective agent plotted in orange and an agent without higher level contextual and affective states plotted in gray so there's it's um i mean one could look at figure 9 a little more closely to see the difference but yes the expectations in like just looking at the top one the expected food location they both converge to left and the per the orange goes a little bit higher so it learns a little faster it's like yeah i'm doing really well on this test this is an easy course then the green line happens and the gray one slowly updates it's like hey i've been right for you know 20 and i was only wrong twice so i'm at 90 percent what's the issue and the orange one's like whoa something's wrong because i'm getting error signals when i wasn't so and the green line here represents the food changing right like that's the context switch i just wanted to contextualize yes yes the context reversal so it was on the left and now it's on the right aka i got a shock and so then also look at this the affective charge the orange one anxiety drop uh blissful ignorance anxiety drop and then the gray one doesn't have affect so we don't see it there but then um yeah or the gray ones uh affective charge is actually a little bit all over the place but we can ask them what that uh means and uh so yeah pretty it's it is super interesting because like learning you know even organic chemistry is an emotional learning process so how actually affect and policy and mental effort and all these things come together it's kind of cool i wonder if this is like just like to take it to the like meta level like is it like um you know self-reflection right like what at what level of integration is that right like so you do the action you select the policy and i think that my doing team mazes don't self-reflect but like when you integrate at that level there's agents that do humans that do self-reflection and then humans that just kind of don't think about it right and so what what is the better adaptation there it just it provides evidence for that i think yep and similarly it's like you know i want to be doing the right thing but then will that preclude you from having a thought that where that's actually being um that expectation is being violated especially if there's sensory evidence that that's the case those are really interesting situations right so like psychopathology like are they just missing some higher level integration like for to lacking that affect you know well ask the authors nice so then um just the additional info section of the paper as we kind of close out in the last little few slides here there are four appendices and the code is available on casper hesp's github and so you can see just from the top lines of the main matlab script yeah it's in matlab it uses markov decision processes and it involves active inference and learning using variational base the factorized version in the spm toolkit using matlab so check out the model streams and let us know how it goes to work with the code we kind of just had one or two more random areas to cover a few directions that this takes us in which we've discussed a little bit one of them is more advanced simulations so can we have two dimensions of affect or could we have another summary variable that represents some other feature of systems could we bring it more into alignment with computational psychiatry and translational research and could we embody robots that implement these kinds algorithms how would that change some type of situational like physical robot or even a software robot if it had these kinds estimators internally and then we're always just thinking about where does active inference come into play which is easy for this paper because that's the whole model but um where are the next empirical steps which they mentioned in 5.7 a little bit and how does this relate to this realism versus instrumentalism question we're always getting at like is this what the systems are or is this how we're gonna model the systems or are those one in the same thing um any broad ideas and questions before this little freudian interlude yeah okay so thanks dave for pointing out um a lot of the recent work of mark solms specifically this new project for a scientific psychology general scheme which the link's given down below but it's um pretty interesting work that has to do integrating recent developments in neuroscience with some freudian and uh ideas and beyond and i guess i just like ideas that aren't named after people because then it gets pretty much like oh who's doing darwinian evolution what does it really mean to be freudian it gets a little bit historiographical instead of about the ideas but the mappings that are made are clear and stand on their own and and put the sort of history hopefully um to the side when we see that there are commonalities across very disparate models and here we see a familiar structure which is kind of like that markov blanket separating the internal and the external states but the letters are a little bit different and so there's an abbreviations table given in this paper and also for many years fristen and carhara harris and others have been writing papers that are about different freudian concepts and relating it to free energy and drive a lot of other points of contact with different psychological threads it's just not uh an area i'm super familiar with with the literature and the debates around but it's an area that's obviously important to many and influences a lot of areas and it's just interesting to see how these developments play into the absolutely still open for evolution field of psychology psychiatry so that was uh pretty much what we wanted to cover in 19 so that was pretty fun and a little little sprint to get the paper finished because it was somewhat of a long paper but it's really worth it so blue any final thoughts or what what are you taking moving forward or what do you hope that we discuss in the 0.1.2 i don't know i'm looking to really discussing uh the bayesian brain integration or or like correlation here i really think that the you know ability to update your beliefs based on what your previous um experiences is like this is like the perfect like case study for this kind of philosophy and i know like the bayesian brain catches some slack sometimes and bayesian statistics catch a lot of slack but in reality like we aren't going to go to the shock side of the box if we know that the food is always in the left side of the box like so having that that update so i'm looking forward to just see what what people have to say about the bayesian brain aspect of this paper if anything we get that far if we can make it all the way through the model right nice yeah i agree i was also curious about okay well what if we could just associate a like you know like click share whatever on social media and then maybe you don't even need the shock or maybe you could add that or something but with that matrix could there be upvote downvote systems where maybe the data already exists and then the positions that somebody could be in would be like i have not seen it or i have seen it something like that they're moving around pages seeing or not seeing things and that's influencing their beliefs about the world and their beliefs about themselves and how how accurate am i if this is what the info i'm getting is and then that's without even going into the instrumentalism realism at least it's something that it feels like to be searching for information or a question that can be useful so it's pretty cool stuff thanks a lot blue and thanks also stephen for joining for 19.0 we'll uh see everyone for 19.1 and two thank you